

# Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-II) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

## 1. Introduction

7 hrs · asked in **all 41 papers** · 6 syllabus sub-topics · **337 marks, 9.9% of the archive** · **all theory, no numerical marks.**

Read this first:

- **Every paper opens here.** Q1 is nearly always a two- or three-part Introduction question for 7 or 8 marks, and it is the cheapest 8 marks in the subject.
- Four questions carry the whole chapter:
  - **Turing test** **TOP 10/41** – what it is, when it is passed, what a machine needs to pass it.
  - **Applications of AI** **TOP 8/41** – almost always the tail of the Turing-test question, never a question on its own.
  - **Intelligent agents** **HOT 7/41** – define, then the types, with diagrams.
  - **PEAS** **HOT 7/41** – write the four rows for a named system.
- **The one idea everything hangs off is the four-approaches grid** (§1.1). Thinking Vs acting, humanly Vs rationally. Every *define AI* answer is that grid; the Turing test is the *acting humanly* box; the rational agent is the *acting rationally* box. Learn the grid and three separate questions collapse into one.
- Pairs the examiner reliably asks together: **Turing test** + **applications** (7 papers), **intelligent agent** + **PEAS** (6 papers), **knowledge** + **learning** (5 papers).
- **PEAS is worth memorising cold.** It is asked in 9 papers for a *named* system (taxi, medical diagnosis, self-driving car, robot in a manufacturing plant, medicine-delivery drone) and no lecture deck works a full example. Four rows, four columns, §1.4.
- Do not write history as prose. The examiner wants **year** → **event** lines, and one of those questions asks specifically which period was the “AI research failure” and why.

### 1.1 What Artificial Intelligence is

Define Artificial Intelligence. Describe AI and its subfields **1 2 4**

**TOP 10/41** 70 Bh, 69 Po, 75 Ba, **77 Ch**, 78 Ba, 79 Ash, 79 Bh, 81 Ba, **77 Ch**, **76 Bh** · opens the Turing-test question in almost every paper, so it is never worth more than 2 marks on its own, and it is never skippable

#### Definitions worth quoting

- **Elaine Rich:** the study of how to make computers do things at which, at the moment, people are better.
  - Best one to write: it explains why the field keeps moving. Once machines win at a task, it stops counting as AI.
- **Kurzweil:** the art of creating machines that perform functions requiring intelligence when performed by people.
- **Minsky:** the science of making computers do things that require intelligence when done by humans.
- **Working definition:** a branch of computer science that builds machines which solve complex problems in a human-like fashion.

#### Intelligence, unpacked

- Ability to **understand**, to **reason**, to **learn**, to **create**.
- No machine has all four. That gap is the whole subject.

#### Intelligent behaviour: three task classes

- **Everyday tasks:** recognise a friend, recognise a caller's voice, translate a language, interpret a photograph, talk, cook a meal.
  - Easiest for people, **hardest for machines**. Worth saying: this is Moravec's paradox.
- **Formal tasks:** prove a theorem, do calculus, play chess, checkers, Go.
- **Expert tasks:** engineering design, medical diagnosis, financial analysis.
  - Hardest for people, **easiest for machines**, because the knowledge is explicit and narrow.

#### Subfields of AI

- Search and problem solving · knowledge representation and reasoning · planning · machine learning · neural networks · expert systems · natural language processing · machine vision and image processing · robotics · fuzzy logic and evolutionary computing.

## The four approaches to AI (the grid everything sits on) 4 7 8

74 Bh, 76 Ba, 82 Bh · asked as “justify that systems that think/act rationally are part of AI” and as “define AI from the dimension of behavioural process and thought process”

|                 | Humanly (measured against people)                                                                                                                                                                                                                                                                                                                                  | Rationally (measured against an ideal)                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thinking</b> | <p><b>Thinking humanly</b> – the cognitive modelling approach. Build a precise theory of how the human mind works, then write it as a program. Needs psychology and introspection. Eg General Problem Solver, Newell and Simon, 1961.</p> <p><b>Criticism:</b> we have no scientific theory of the brain's internal activity, so the model cannot be verified.</p> | <p><b>Thinking rationally</b> – the laws of thought approach. Encode correct reasoning as formal logic, then let an inference engine derive conclusions. Eg “Socrates is a man; all men are mortal; therefore Socrates is mortal.”</p> <p><b>Criticism:</b> informal, uncertain knowledge is very hard to state in logical notation, and solving a problem in principle is not solving it in practice.</p> |
| <b>Acting</b>   | <p><b>Acting humanly</b> – the <b>Turing test</b> approach. Model what people do, not how they think. Judged purely on external behaviour. Eg ELIZA.</p> <p>This is the box §1.2 lives in.</p>                                                                                                                                                                     | <p><b>Acting rationally</b> – the <b>rational agent</b> approach. Do whatever is expected to achieve the best outcome. Covers reflex action, which needs no reasoning at all.</p> <p><b>The approach modern AI actually uses</b>, because it is general and it can be measured. This is the box §1.3 lives in.</p>                                                                                         |

### How to answer “justify that X is part of AI”

- Name the box, give its yardstick, give an example system, give its criticism.
- **Thinking rationally** is part of AI because logic-based inference (FOPL, resolution, Ch 4) derives new true facts from known ones without any human in the loop.
- **Acting rationally** is part of AI because an agent that maximises its performance measure in an uncertain environment is doing what intelligence is *for*, whether or not it reasons the way people do.
- **Say which is preferred and why:** acting rationally, because rationality is mathematically well defined and generalises, while “humanly” needs a theory of mind we do not have.

### Types of AI by capability

- **Artificial Narrow Intelligence (ANI, weak AI):** one task or one narrow domain. Everything that exists today. Eg spam filter, chess engine, a speech recogniser.
- **Artificial General Intelligence (AGI, strong AI):** any intellectual task a human can do, transferring skill between domains. Does not exist.
- **Artificial Super Intelligence (ASI):** exceeds human ability in every field including creativity and social skill. Hypothetical.

### Weak Vs strong AI, the philosophical sense

- **Weak:** the machine *simulates* thinking.
- **Strong:** the machine genuinely *has* a mind.
- Searle's Chinese Room argues strong AI is impossible: symbol manipulation is not understanding. verified/added

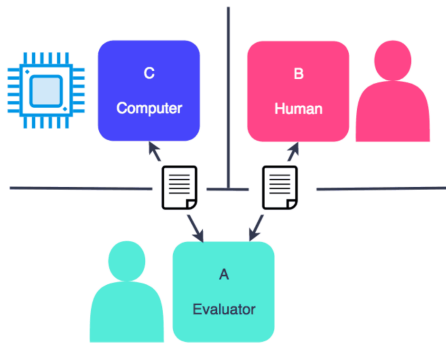
## 1.2 The Turing test

### What is the Turing test? When is a machine said to have passed it? 2 4 5 7

TOP 10/41 70 Bh 5, 69 Po 7, 75 Ba 2, 77 Ch 3, 78 Ba 2, 79 Ash 4, 79 Bh 5, 81 Ba 6, 77 Ch 4, 76 Bh 3 · the single most-asked question in the subject's first chapter

#### The setup

- Proposed by **Alan Turing**, 1950, in *Computing Machinery and Intelligence*, to replace the unanswerable “can machines think?” with a testable question, “can machines behave intelligently?”
- Three rooms, no sight or sound between them:
  - a human **interrogator**,
  - a human **respondent**,
  - the **machine**.
- The interrogator questions both over a text-only channel (a teletype, so voice and handwriting give nothing away) and must say which is which.



**Draw this: the arrangement is the definition.** A is the interrogator, B the human respondent, C the machine. The wall and the text-only channel are the two things that must appear – they are what stops the test measuring anything except behaviour. A knows one of B, C is a machine and must say which.

#### When it is passed

- The machine passes when the interrogator **cannot do better than chance**, i.e. is wrong about half the time.
- Turing’s own operational figure: an average interrogator has **under a 70 % chance** of identifying correctly after **5 minutes** of questioning, so the machine fools them **30 %** of the time.
- **Write both:** the half-the-time principle, then the 5-minute / 30 % figure as the concrete criterion.

#### The four capabilities a machine needs to pass

- **Natural Language Processing** → to communicate in English at all.
- **Knowledge Representation** → to store what it knows and what it is told during the interview.
- **Automated Reasoning** → to answer questions and draw new conclusions from what is stored.
- **Machine Learning** → to adapt to new circumstances and detect patterns it was not given.

#### Total Turing test adds two more

- The plain test deliberately avoids physical contact, so it never tests perception or manipulation. The **total Turing test** puts a video link and a hatch for passing objects into the setup, adding:
  - **Computer vision** → to perceive objects,
  - **Robotics** → to manipulate them and move about.
- **6 capabilities in total.** (77 Ch and 76 Bh ask for the total test by name)

## ELIZA, the standard example

- Written at MIT, **1964–66**, by **Joseph Weizenbaum**. First script **DOCTOR**, imitating a Rogerian psychotherapist.
- Mechanism: a table of **syntactic patterns**, each with a canned reply that echoes part of the input back after simple substitutions (*my* → *your*). **No understanding, no knowledge base, no memory.**
- Weizenbaum was disturbed by the reaction: psychiatrists thought it clinically useful and users confided in it.
- **The exam point:** ELIZA shows a machine can be *taken* for a human without any intelligence behind it, which is the standing objection to the test itself.

## Importance of the Turing test

- Turns a philosophical question into an **operational, repeatable** one.
- Sets a **behavioural** yardstick: judge output, not internal mechanism, so it does not need a theory of mind.
- Defines the **research agenda**: its four required capabilities are four whole subfields of AI, and this syllabus is mostly those four.
- Still the benchmark by which conversational systems are reported.

## Limitations

- Rewards **deception**, not competence. Passing needs human-like error and delay, so a machine that answers arithmetic instantly and correctly *fails*.
- **Not reproducible** and not mathematically definable; the result depends on the interrogator.
- Tests only what people happen to do, so it caps machine intelligence at human behaviour rather than measuring intelligence itself.

## If the Turing test is passed, does that show computers exhibit intelligence? 2 3 4 7

**HOT 4/41** 72 Ma 7, 73 Bh 7, 81 Bh 4+3, 80 Ch 2+4 · **an argue-both-sides question**: take a side, but the marks are in giving both cases first. 81 Bh and 80 Ch tack the *reverse* Turing test on the end

### Case for “yes”

- Intelligence has **no agreed internal definition**, so behaviour is the only thing that can be measured at all.
- We attribute intelligence to *other people* on exactly this evidence, and nothing else. Refusing the machine the same inference is inconsistent.
- Passing is not a trick: it requires NLP, knowledge, reasoning *and* learning working together, which is a real and general capability.

### Case for “no” (the stronger answer)

- **Imitation is not cognition.** ELIZA fooled people with pattern substitution and knew nothing.
- **Searle’s Chinese Room**: a person following symbol-manipulation rules can answer Chinese questions perfectly while understanding no Chinese. Syntax does not give semantics. **verified/added**
- Tests only **one narrow channel**, text conversation. Says nothing about perception, physical skill, creativity or goals.
- **Recommended stance**: passing shows *intelligent behaviour*, not *intelligence*. It is a sufficient test of appearing intelligent and a necessary-but-not-sufficient test of being so.

## Reverse Turing test 3 4

**verified/added** 81 Bh 3, 80 Ch 4 · 2 papers, both under 6 marks, so no tier chip · **not in any lecture deck in this folder**, researched and added, and it has been asked twice in the last three years

### What it is

- The roles are swapped: **a machine interrogates, and a human must prove it is human.**
- The test succeeds when the machine can tell human from machine, which is the opposite goal to Turing’s.
- **CAPTCHA** is the everyday instance: *Completely Automated Public Turing test to tell Computers and Humans Apart*. A distorted-text or image-grid challenge that people pass easily and programs do not.

### Turing Vs reverse Turing, side by side

- **Who judges**: human, Vs machine.
- **Who is on trial**: machine, Vs human.
- **Passed when**: the judge cannot tell them apart, Vs the judge *can*.
- **Purpose**: measure machine intelligence, Vs block bots from a web service.
- **Note the irony worth a mark**: every CAPTCHA solved by a program is a small Turing test passed, which is why CAPTCHAs keep getting harder.

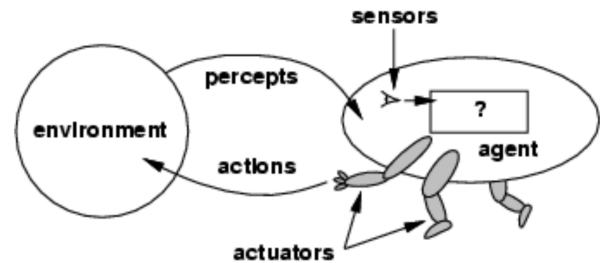
### 1.3 Intelligent agents

## What is an intelligent agent? Explain the different types with examples 1 2 6 8

**HOT 7/41** 74 Bh 1+6, 80 Bh 2+6, 80 Ash 2+6, 79 Jth 1+6, 75 Bh 8, 76 Ba 8, 71 Bh 4+4 · 75 Bh asks specifically **how a learning agent works**, so do not stop at the four standard types

#### Definitions to write first

- **Agent:** anything that perceives its environment through **sensors** and acts upon that environment through **actuators**.
  - **Human agent:** eyes and ears as sensors; hands, legs and mouth as actuators.
  - **Robotic agent:** cameras and infrared range finders as sensors; motors as actuators.
  - **Software agent:** keystrokes and file contents as sensors; screen writes and network packets as actuators.
- **Percept:** what the agent perceives at one instant. **Percept sequence:** everything it has ever perceived.
- **Agent function** maps percept sequence to action,  $f : P^* \rightarrow A$ .
- **Agent program** is the concrete implementation that runs on the hardware.
- **Agent = architecture + program.** One line, and it is often a mark on its own.



The agent, its sensors and actuators, and the environment loop.

**Draw this first** in any agent answer; every later diagram is this one with a box added inside the agent.

#### Rational agent

- For each possible percept sequence, a rational agent selects the action expected to **maximise its performance measure**, given the evidence in the percept sequence and whatever built-in knowledge it has.
- **Performance measure:** the objective criterion for success, fixed by the designer, not the agent. Eg for a vacuum cleaner: dirt collected, time taken, electricity used, noise made.
- **Rationality is not omniscience.** An omniscient agent knows the actual outcome of its actions; that is impossible. Rationality maximises *expected* performance given what is knowable.
- **Autonomy:** an agent is autonomous when its behaviour is determined by its own percepts and experience rather than only by built-in knowledge. Autonomy requires learning.
- **Information gathering** and **exploration** are rational acts: it can be rational to act now purely to get a better percept later.

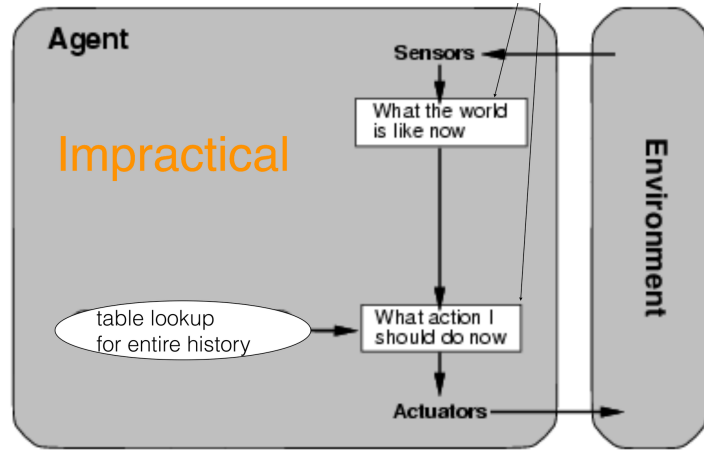
#### Environment properties (worth 2 marks when asked)

- **Fully Vs partially observable:** can the sensors see the whole relevant state?
- **Deterministic Vs stochastic:** does the next state follow from the current state and action alone?
- **Episodic Vs sequential:** does the current action affect later ones?
- **Static Vs dynamic:** does the world change while the agent deliberates?
- **Discrete Vs continuous:** countable states and actions, or not?
- **Single agent Vs multi-agent:** is anything else optimising against it?
- Eg a taxi driver's world is partially observable, stochastic, sequential, dynamic, continuous and multi-agent. **The hardest kind on every axis.**

#### The five agent types, in increasing generality

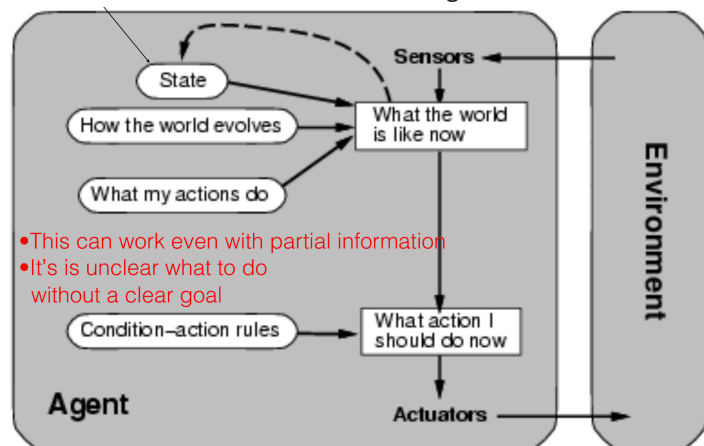
### 1. Table-driven agent

- Holds a lookup table indexed by the *entire percept history*, and reads the next action out of it.
- **Impractical**: the table grows exponentially with the length of the history. A taxi with a camera would need more entries than there are atoms in the universe.
- Included only to show why the later types exist.



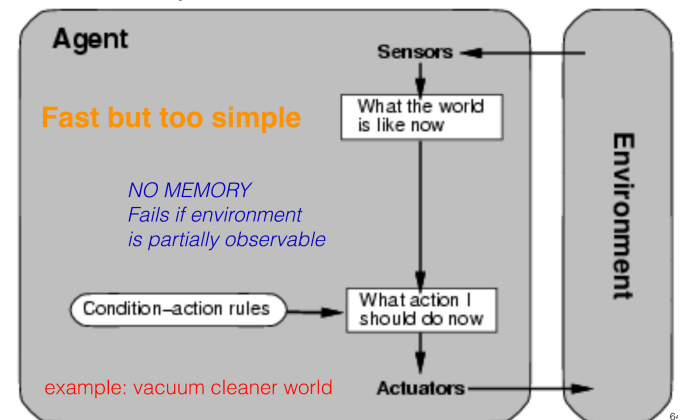
### 3. Model-based reflex agent

- Keeps **internal state** = the part of the world it cannot currently see.
- Needs two pieces of knowledge, together called the **model of the world**:
  - how the world evolves on its own,
  - what my actions do to it.
- **Fixes partial observability**. Eg a taxi remembering which cars are in the lanes it cannot see right now.



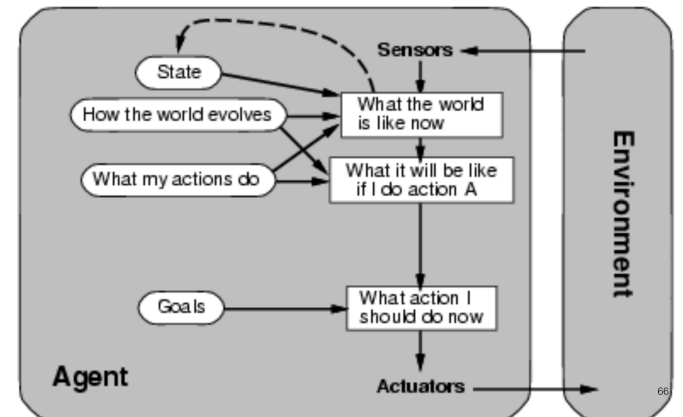
### 2. Simple reflex agent

- Acts on the **current percept only**, through **condition-action (if-then) rules**. Implemented as a production system (Ch 2).
- **Stateless**: no memory of past world states.
- Eg
  - if car-in-front-is-braking then initiate-braking;
  - the vacuum-cleaner agent that sucks whenever the current square is dirty.
- **Fails when the environment is partially observable**, and can loop forever in that case.



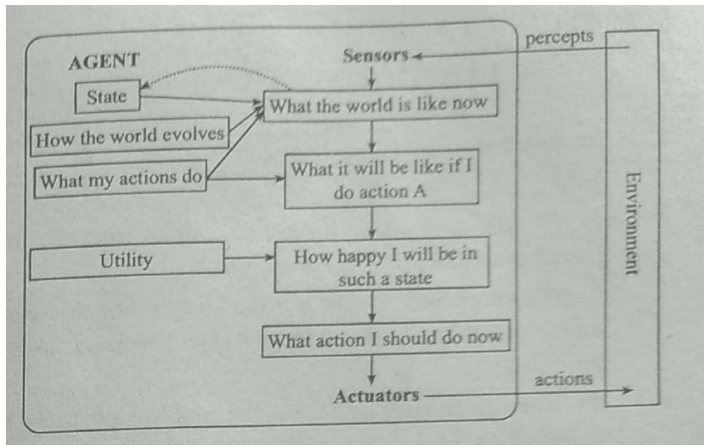
### 4. Goal-based agent

- Adds **goal** information describing desirable situations, and asks "what will it be like if I do action A?"
- Considers the **future**, so it can plan and search (Ch 3), not just react.
- **Far more flexible**: change the goal and the behaviour changes, with no rules rewritten. A reflex agent needs every rule rewritten.
- Eg a taxi told to go to a different destination.



## 5. Utility-based agent

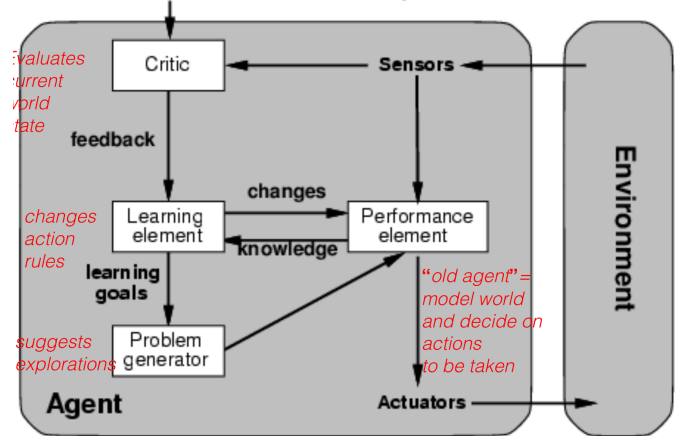
- A goal is binary: reached or not. A **utility function** maps a state to a number saying **how happy** the agent is there.
- Needed for two cases a goal cannot settle:
  - **conflicting goals** – fast Vs safe Vs cheap: utility trades them off;
  - **uncertain goals** – pick the action with the highest *expected* utility.
- Eg a taxi choosing between a quick risky route and a slow safe one.



## Learning agent (asked by name, 75 Bh 8)

- Any of the four above, wrapped in machinery that **improves it with experience**. Four parts:
  - **Performance element** – the old agent: takes percepts, chooses actions.
  - **Critic** – judges the outcome against a fixed **performance standard** and reports feedback. The standard must be external, or the agent could just lower its own bar.
  - **Learning element** – uses that feedback to change the performance element.
  - **Problem generator** – deliberately suggests **exploratory** actions that look suboptimal now, so the agent discovers better behaviour later.
- **The exam point:** learning is what makes an agent truly autonomous, which is the link straight to §1.9 and to Ch 6.

Performance standard *better modeling, new action rules, etc.*



## 1.4 PEAS: specifying the task environment

### Define PEAS. Give the PEAS information for a named system 2 3 5 6

**HOT 7/41** 80 Ba (medical diagnosis) 3, 78 Ch 4, 74 Bh 3, 81 Ash (taxi + robot) 5, 82 Ka (robot) 3, 82 Ba (self-driving car) 6, 82 Bh (medicine-delivery drone) 5 · **asked with a different system every single time**, so learn the four rows, not four memorised answers

#### What and why

- Before designing an agent, its **task environment** must be specified. PEAS is that specification:
  - P – Performance measure:** what counts as success. Must be about the *environment*, not the agent's internal state.
  - E – Environment:** what the agent operates in and on.
  - A – Actuators:** how it acts.
  - S – Sensors:** how it perceives.
- Why it must come first:** the performance measure decides what “rational” means, and the sensors and actuators decide what the agent can even represent. Design before PEAS is design without a specification.

#### Three mistakes that cost marks

- Writing the performance measure as “*work well*”. Give measurable terms: safe, fast, legal, comfortable, maximise profit.
- Putting an actuator in the sensor row. A horn is an actuator; a microphone is a sensor.
- Forgetting that a **display** is an actuator when the agent uses it to tell a person something.

#### Self-driving car (82 Ba 6)

- P:** reach the destination safely, obey traffic law, minimise journey time and fuel, ride comfort, zero collisions.
- E:** roads and lane markings, other vehicles, pedestrians and cyclists, traffic signals, road signs, weather, the passenger.
- A:** steering, throttle, brakes, gear selection, indicators, horn, headlights, passenger display and speaker.
- S:** cameras, LIDAR, radar, ultrasonic parking sensors, GPS, IMU and accelerometer, wheel-speed sensors, microphone.

#### Ready-made answers for the systems actually asked

- Taxi driver** (81 Ash)
  - P:** safe, fast, legal, comfortable trip, maximise profits.
  - E:** roads, other traffic, pedestrians, customers, weather.
  - A:** steering, accelerator, brake, indicator, horn, display.
  - S:** cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard.
- Medical diagnosis system** (80 Ba)
  - P:** healthy patient, minimise costs and lawsuits.
  - E:** patient, hospital, staff.
  - A:** screen display of questions, tests, diagnoses, treatments, referrals.
  - S:** keyboard entry of symptoms, findings and the patient's answers.
- Part-picking robot in a plant** (81 Ash, 82 Ka)
  - P:** percentage of parts placed in the correct bin.
  - E:** conveyor belt with parts, the bins.
  - A:** jointed arm, gripper.
  - S:** camera, joint angle sensors.

#### Medicine-delivery drone (82 Bh 5) verified/added

- No source note covers this one;** state your assumptions, as the paper explicitly allows, then give four clean rows.
- P:** medicine delivered intact, to the right address, within the time window, maximum payload per charge, no collision, no airspace violation.
- E:** air corridor, buildings, terrain and hills, trees and power lines, weather and wind, other drones and aircraft, the recipient, the dispatch centre.
- A:** rotors and motors, payload release hatch, landing gear, status lights, radio downlink to the dispatch centre.
- S:** GPS, camera, LIDAR or ultrasonic altimeter, IMU and gyroscope, barometer, wind and temperature sensors, battery gauge, radio uplink.

## 1.5 Importance and limitations of AI; AI Vs human intelligence

### Differentiate natural and artificial intelligence. Can AI replace humans? 3 4 5 7

PIN 3/41 79 Ch 1+3+4, 78 Po 3+5, 73 Bh 7 · growth areas asked at 71 Bh 4, disadvantages at 78 Ba 2

|             | Human intelligence                            | Artificial intelligence                       |
|-------------|-----------------------------------------------|-----------------------------------------------|
| Origin      | Biological, inherited and grown               | Engineered, programmed and trained            |
| Learns from | A few examples, plus intuition and context    | Very large data sets, no context of its own   |
| Speed       | Slow, tires, needs rest                       | Constant, no fatigue, runs continuously       |
| Accuracy    | Errs when tired or emotional                  | Exact and repeatable within its domain        |
| Breadth     | <b>General:</b> transfers skill to a new task | <b>Narrow:</b> fails outside the trained task |
| Creativity  | Original ideas, imagination                   | Recombines what it was trained on             |
| Emotion     | Empathy, judgement, ethics                    | None; only what is coded as a measure         |
| Adapting    | Handles the wholly unfamiliar                 | Needs retraining or reprogramming             |
| Cost        | Salary, training time                         | High to build, near-zero to copy              |

#### Importance and advantages of AI

- **Automates repetitive work** → people move to judgement work.
- **Better decisions:** finds patterns in data volumes no person can read.
- **Available always:** no fatigue, no shift limits, consistent output.
- **Works where people cannot:** deep sea, space, reactor cores, disaster zones, mine clearance.
- **Fewer human errors** in exact, repetitive tasks.
- **Reaches scarce expertise:** an expert system puts specialist knowledge in a village clinic.
- **Speeds research:** drug discovery, protein folding, materials.

#### Growth areas (71 Bh)

- Machine learning and deep learning · NLP and speech · computer vision · robotics and autonomous vehicles · expert and decision-support systems · recommender systems · medical imaging and diagnosis.

#### Can AI replace humans? Answer: in tasks, yes; in roles, no

- **Where it replaces:** repetitive, rule-bound, data-heavy work. Data entry, first-line support, defect inspection, routine diagnosis, translation drafts.
- **Where it cannot, and why:**
  - **No common sense** outside the training distribution.
  - **No moral or legal responsibility;** someone must be accountable.
  - **No genuine creativity or empathy,** both needed in teaching, care, negotiation, design.
  - **Depends on data** that people must collect, label and correct.
- **Conclusion to write:** AI is a tool that shifts human work up the skill ladder, not a replacement for the worker. The realistic risk is *displacement of specific tasks*, not obsolescence of people.

#### Disadvantages and limitations (78 Ba 2)

- **Cost:** high to build, and needs specialists to maintain.
- **No common sense:** fails absurdly on inputs slightly outside its training.
- **No creativity:** recombines, does not originate.
- **Job displacement** in routine occupations.
- **Opaque:** a neural network cannot explain its decision, which is unacceptable in medicine and law.
- **Inherits bias** from its training data, then applies it at scale.
- **No ethics or accountability,** so misuse is easy: surveillance, autonomous weapons, deepfakes.
- **Data-hungry and energy-hungry.**

#### Two named ethical positions worth a line

- **Weizenbaum (1976):** a real AI would be autonomous and would not share our motives or ethics, so the research is unethical.
- **Asimov's three laws of robotics:** do not harm a human or by inaction allow harm; obey humans; protect itself. Precedence runs lower number first.

## 1.6 AI and related fields

### Describe AI and its subfields; which fields contributed to AI? 4

76 Bh 4 · asked once, but it is a two-minute answer and it recurs as a sub-part of "define AI"

#### What each field gave AI

- **Philosophy:** logic, the idea of the mind as a physical system, foundations of learning, language and rationality.
- **Mathematics:** formal representation and proof, algorithms, computation, decidability and intractability, probability, statistics, linear algebra.
- **Psychology:** perception, motor control, adaptation, and the experimental methods behind cognitive modelling.
- **Economics:** formal theory of rational decision, utility, game theory.
  - This is where the *utility* in a utility-based agent comes from.
- **Linguistics:** grammar, syntax and knowledge representation. NLP sits directly on it.
- **Neuroscience:** the physical substrate of mental activity. Neural networks are a deliberate, loose imitation of it.
- **Control theory:** stability, feedback, optimal agent design. The agent-environment loop is a control loop.
- **Computer engineering:** the machines fast enough to run any of it.

#### Fields AI overlaps with, and how it differs

- **Machine learning** is a subfield of AI, not a synonym: AI also covers search, logic and planning, which do not learn at all.
- **Data mining** (Ch 6 and the Data Mining elective) shares algorithms but aims at *discovering patterns in stored data*; AI aims at *acting well*.
- **Robotics** supplies the body; AI supplies the decision making.
- **Cognitive science** is the joint enterprise of AI and psychology: AI wants machines that work, cognitive science wants a theory of the mind.
- **Operations research** shares optimisation and search but assumes a fully specified model.

#### Programming languages of AI

- **LISP:** McCarthy, late 1950s. Functional, symbolic, interactive.
- **PROLOG:** France 1970, developed at Edinburgh 1975–79. Logic programming: a program is facts + rules, and running it is inference. **The syllabus lab is in one of these two.**

## 1.7 Brief history of AI

### Discuss the brief history of AI with chronological development 2 5 6 7

**PIN 3/41** 70 Ma 2+6, 71 Bh 2+5, 72 Ma 7 · 72 Ma asks specifically **which period was the “AI research failure” period and why**, so learn the two AI winters, not just the timeline

#### The founding era, 1943–1979

| Year    | Event                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1943    | <b>McCulloch and Pitts</b> : Boolean circuit model of the neuron. The first neural network.                                                  |
| 1950    | <b>Turing</b> , <i>Computing Machinery and Intelligence</i> . The Turing test.                                                               |
| 1956    | <b>Dartmouth meeting</b> : the term “Artificial Intelligence” is adopted. <b>The birth of the field.</b>                                     |
| 1950s   | Samuel’s checkers program, which learned and beat its author; Newell and Simon’s Logic Theorist; Gelernter’s Geometry Theorem Prover (1959). |
| 1957    | <b>Rosenblatt’s Perceptron.</b>                                                                                                              |
| 1958    | <b>McCarthy</b> defines <b>LISP</b> and describes the Advice Taker.                                                                          |
| 1963    | McCarthy starts the Stanford AI lab. Slagle’s SAINT solves first-year calculus.                                                              |
| 1965    | <b>Robinson’s resolution</b> : a complete theorem-proving algorithm for first-order logic. Ch4 is built on it.                               |
| 1966    | <b>ELIZA</b> , Weizenbaum.                                                                                                                   |
| 1969–79 | Knowledge-based and <b>expert systems</b> : <b>DENDRAL</b> , <b>MYCIN</b> , <b>SHRDLU</b> . Minsky’s <b>frames</b> (1975).                   |

#### The “AI research failure” period

- Two collapses in funding and confidence, both called **AI winters**. The question almost always means the first.
- **First AI winter, 1966–73**, the “dose of reality”. In 1957 Simon predicted a computer chess champion and a major theorem proved within **10 years**; both took roughly **40**. Three reasons it failed:
  - **No subject knowledge**. Early programs succeeded by *syntactic manipulation* alone. Machine translation is the standard example: word-by-word substitution with no meaning behind it.
  - **Intractability**. Programs solved toy problems by trying combinations of steps. Scaling the problem exploded the search, and the 1965 discovery of NP-completeness explained why faster hardware could not fix it.
  - **Structural limits**. Minsky and Papert’s *Perceptrons* (1969) proved a single-layer perceptron cannot represent XOR, and neural network research nearly stopped for fifteen years. **This is the same XOR result you prove in Ch 7.**

#### Industry, science and the modern era, 1980–

| Year  | Event                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------|
| 1980  | AI becomes an <b>industry</b> . XCON saves DEC an estimated \$40 million a year.                           |
| 1986  | <b>Neural networks return</b> , on back-propagation.                                                       |
| 1987  | AI becomes a <b>science</b> : results measured against benchmarks, and statistics and probability adopted. |
| 1995  | The <b>intelligent agent</b> becomes the unifying idea of the field.                                       |
| 1997  | <b>Deep Blue</b> beats Kasparov at chess.                                                                  |
| 2011  | IBM Watson wins <i>Jeopardy!</i>                                                                           |
| 2016  | <b>AlphaGo</b> beats Lee Sedol at Go.                                                                      |
| 2020– | Large language models, <b>GPT-3</b> onward.                                                                |

- **Write it as a table, not as prose.** Year, then event, then one clause on why it mattered. Dates with no significance score badly.

#### The second winter, and what ended both

- **Second AI winter, late 1980s to early 1990s**: the expert-system market collapsed. The systems were **brittle** outside their narrow domain, **could not learn**, and cost more to maintain than they saved.
- **What ended each**:
  - the first ended with **knowledge-based systems** – stop searching blindly and put the domain knowledge in;
  - the second ended with **statistical machine learning**, then cheap parallel hardware and large data sets.
- **The pattern behind both**, and the sentence that earns the last mark: each winter followed a period of **over-promising on a technique that did not scale**, and each ended when the field adopted a *different* technique rather than more of the same.

## 1.8 Applications of AI

### Discuss fields of daily life where AI has been applied 2 4 6 7

**TOP 8/41** 69 Bh 7, 72 Ash 7, 73 Ma 6, 81 Ch 2+4, 75 Ba, 77 Ch, 78 Ba, 78 Po · usually the tail of a Turing-test or define-AI question, which is why it is worth 4 marks and two minutes. Most papers ask for **any two**, explained

#### How to answer for full marks

- The question says *explain two fields*, not *list ten*. A list scores badly.
- For each field give: **the task**, **the AI technique behind it**, and **a named example**. Three sentences each is the whole answer.
- 81 Ch asks for applications **as a problem-solving tool**, so frame each one as “the problem, and how AI solves it”.

#### Two safe choices, worked

- **Healthcare**. Problem: expert diagnosis is scarce and slow. Technique: neural networks over medical images (Ch 7), plus rule-based expert systems for diagnosis. Examples: tumour detection on CT and MRI, IBM Watson for Oncology, diabetic-retinopathy screening from retinal photographs.
- **Transport**. Problem: driving needs continuous perception and decision under uncertainty. Technique: computer vision and sensor fusion feeding a utility-based agent, plus A\* and its variants for route search (Ch 3). Examples: self-driving cars, Google Maps ETA prediction, Pathao and inDrive dispatch and pricing.

#### The wider list, grouped

- **Everyday software**: search-engine ranking and autocomplete, social-media feeds, Netflix and YouTube recommendation, spam filters, photo tagging.
- **Language**: machine translation, speech recognition, chatbots and virtual assistants, Nepali NLP tools (Ch 7).
- **Finance**: fraud and money-laundering detection, credit scoring, algorithmic trading.
- **Industry and agriculture**: robotic assembly and part picking, predictive maintenance, visual quality inspection, crop-disease detection from leaf images, yield prediction.
- **Games**: Deep Blue, AlphaGo, and the bots in Dota 2 and PUBG.
- **Science and space**: astronomical object classification, protein folding, space-mission monitoring, drug discovery.
- **Public services**: weather and flood forecasting, traffic signal control, disaster response, biometric identification.

#### Nepal-specific examples, if the question invites them

- Pathao and inDrive dispatch · eSewa and Khalti fraud detection · Nepali speech and OCR · landslide and flood early warning · crop-disease apps.

## 1.9 Knowledge and learning

### What is knowledge and learning? Why are they important? 2 4 5 8

**HOT 5/41** 73 Bh 4+4 , 74 Ma 2+2+2+4 , 72 Ash 2+2+2+4 , 74 Bh 4+4 , 79 Jth 4+4 · also inside the Turing question at 79 Ash 2+2 and 71 Ma 2  
· **the quotable line the examiner wants back:** "Learning is an essential characteristic for intelligent agents"

|         | Data                            | Information                                   | Knowledge                                             |
|---------|---------------------------------|-----------------------------------------------|-------------------------------------------------------|
| What    | Raw facts, unorganised          | Data processed into a meaningful, usable form | Information organised and understood, ready to act on |
| Form    | Numbers, symbols, a database    | Tables, reports, summaries                    | Rules, procedures, models                             |
| Answers | nothing on its own              | who, what, where, when                        | <b>how and why</b>                                    |
| Example | $V = 12$ , $I = 6$ ,<br>$R = 2$ | $V = I \times R$ holds for these readings     | Use $V = IR$ to design a circuit                      |

- **Intelligence** sits one step above knowledge: the ability to use knowledge to solve new problems and adapt to new situations.
- **Write the chain:** data → information → knowledge → intelligence. Each step adds structure and context.

#### Knowledge, and why AI needs it

- **Knowledge:** the condition of knowing something with familiarity gained through experience or association; what lets us explore and decide.
- Stored by people in **natural language**, by machines as **symbols, strings and structures**.
- Forms it takes: simple facts, relations between concepts, **associations, inheritance hierarchies**, procedures.
- **Importance:** enough knowledge is what produces intelligence. A search algorithm with no domain knowledge is blind (Ch 3); the same algorithm with a heuristic is informed and vastly faster.  
**Knowledge is what turns search into intelligence.**
- **The three designer problems** (they run through Ch 4 and Ch 5):
  - **Knowledge acquisition** – getting it out of the domain and its experts.
  - **Knowledge representation** – encoding it: logic, semantic nets, frames, rules.
  - **Knowledge manipulation** – inferring new conclusions from it.

#### Learning

- **Herbert Simon (1983):** "Learning is the phenomenon of knowledge acquisition in the absence of explicit programming."
- Fuller form: learning denotes **changes in the system that are adaptive**, in the sense that they let it do the same task, or a task drawn from the same population, more efficiently and more effectively next time.
- **Three factors in any learning system:**
  - **Changes** – what in the system is altered, and how it is represented.
  - **Generalisation** – performance must improve on *similar* tasks, not only the one seen. Without this it is memorisation.
  - **Improvement** – the change must not degrade performance; the system must detect and prevent that.
- **Knowledge Vs learning in one line:** knowledge is the *content* an agent holds; learning is the *process* that changes that content.

#### "Learning is an essential characteristic for intelligent agents"

- **Agree, and justify in four steps:**
  - **The designer cannot foresee every situation.** A fixed rule table is only as good as its author's imagination, and the table-driven agent shows how quickly enumeration becomes impossible.
  - **Environments change.** Roads are rebuilt, spam changes its wording, prices move. A non-learning agent's performance decays with no fault of its own.
  - **Autonomy requires it.** An agent whose behaviour comes only from built-in knowledge is running the designer's intelligence, not its own. Learning is exactly what makes it autonomous.
  - **The Turing test demands it.** Machine learning is one of the four capabilities needed to pass, so an agent that cannot learn cannot be judged intelligent on the field's own benchmark.
- **Example to close with:** a spam filter shipped with fixed rules is beaten within weeks; one that learns from each user's marking of messages keeps working. Same architecture, and learning is the only difference.

## Belief, hypothesis and knowledge 4

71 Bh 4 · asked once, as the tail of the growth-areas question. A three-row table is the whole answer

| Term              | Means                                                                                                                | Confidence                           |
|-------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Belief</b>     | Any statement an agent holds to be true. It need not be verified and may turn out false.                             | Subjective, unverified.              |
| <b>Hypothesis</b> | A belief <i>put forward for testing</i> , backed by some evidence and held provisionally until confirmed or refuted. | Justified belief, not yet confirmed. |
| <b>Knowledge</b>  | A belief that is <i>true</i> and <i>justified</i> . Verified, and safe to reason from.                               | Confirmed.                           |

- **The relation, in one line:** knowledge  $\subset$  hypothesis  $\subset$  belief. Every hypothesis is a belief; only a confirmed, true one becomes knowledge.
- **Why AI cares:** an agent must act on beliefs, because it rarely has knowledge. That is precisely why Ch 4 needs probability and belief networks rather than plain logic.

### Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short | Means                                                | Short   | Means                                                                      |
|-------|------------------------------------------------------|---------|----------------------------------------------------------------------------|
| AI    | artificial intelligence                              | ANI     | artificial narrow intelligence                                             |
| AGI   | artificial general intelligence                      | ASI     | artificial super intelligence                                              |
| KR    | knowledge representation                             | KRR     | knowledge representation and reasoning                                     |
| PEAS  | performance measure, environment, actuators, sensors | GPS     | General Problem Solver                                                     |
| NLP   | natural language processing                          | NLU     | natural language understanding                                             |
| NLG   | natural language generation                          | CAPTCHA | completely automated public Turing test to tell computers and humans apart |
| ML    | machine learning                                     | ANN, NN | artificial neural network                                                  |
| MLP   | multilayer perceptron                                | BP      | back-propagation                                                           |
| SOM   | self-organizing map                                  | GA      | genetic algorithm                                                          |
| FIS   | fuzzy inference system                               | ID3     | Iterative Dichotomiser 3                                                   |
| CSP   | constraint satisfaction problem                      | DFS     | depth first search                                                         |
| BFS   | breadth first search                                 | DLS     | depth limited search                                                       |
| IDS   | iterative deepening search                           | UCS     | uniform cost search                                                        |
| CNF   | conjunctive normal form                              | WFF     | well-formed formula                                                        |
| FOL   | first-order logic                                    | FOPL    | first-order predicate logic                                                |
| RRS   | resolution refutation system                         | BBN     | Bayesian belief network                                                    |

| Short   | Means                      | Short  | Means                                                                                       |
|---------|----------------------------|--------|---------------------------------------------------------------------------------------------|
| ES      | expert system              | KB     | knowledge base                                                                              |
| CD      | conceptual dependency      | LISP   | list processing (the language)                                                              |
| MOP     | memory organization packet | isa    | the class-to-class link of a semantic net ( <i>instance</i> is the individual-to-class one) |
| attr(s) | attribute(s)               | repo   | repository                                                                                  |
| freq    | frequency                  | config | configuration                                                                               |
| info    | information                | no.    | number                                                                                      |
| Eg      | example                    | Vs     | compare / differentiate                                                                     |
| w/, w/o | with, without              | +Fig   | draw the figure                                                                             |
| →       | leads to, therefore        | =      | is, equals                                                                                  |

## Reading the year tags and chips

**Exam-paper month codes:** Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

**Bold** year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

**Three course codes, three faces.** Plain = **CT 653** (BCT, III-II, 26 papers) · `monospace` = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

**Tier chips** over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth  $\geq 6$  marks.

**Mark chips** such as **2+6** are the marks that question actually carried. `syllabus, no PYQ yet` marks a syllabus topic never yet examined but written up anyway. `verified/added` marks anything researched and added that no source note in this folder covers.